COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 Data Structures And Algorithms	**SECTION:**	CPE21S2
DATE PERFORMED:	10/14/26	**DATE SUBMITTED:**	10/14/26
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HOA 12.1 Algorithmic Strategies

1. PROCEDURE OUTPUT

ILO A: Demonstrate an understanding of algorithmic strategies (Pre-Lab Review)

Table 12-1. Algorithmic Strategies and Examples

Strategy	Algorithm	Analysis
Recursion	Fibonacci	The function calls itself with n-1 and n-2. It stips when it is 0 or 1, and n gets smaller on every call.
Brute Force	Breadth First Search	The function will check at the first input then check which condition they can fulfill if neither subtract one until n == 1 and check which one has a lower steps value.
Backtracking	Depth First Search	The function will dive on one path first then backtrack to check whether another path has a lower steps value.
Greedy	Selection Short	On each pass it picks the smallest remaining elements and places it, and never goes back to change it
Divide-and-Conquer	Merge Sort	Divide: split the array into two halves. Solve: sort each half recursively. Combine: merge the two sorted halves.

Table 12-2. Memoization Implementation

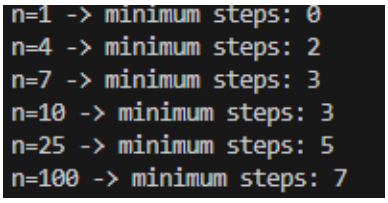
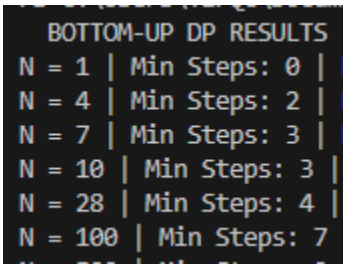
Screenshot	
Analysis	The program saves each answer in the memo array, so every value of n is solved only once. For n = 10 the answer is 3 because 10 -1 = 9 /3 = 3, 3/3 = 1. Which matches the handout and is better than greedy answer of 4 steps.

Table 12-3. Bottom-Up Dynamic Programming Implementation

Screenshot	
Analysis	The program uses a for loop and array filling where it starts at the base case and towards the input n. Where it checks the conditions and if it matches a previous index and has the lowest value(minimum history) it will take its value and + 1 to its value then store it as how many steps it took.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Problem: Count the number of paths in a matrix with a given cost to reach the destination cell

Pseudocode

```
function countPaths(mat, i, j, cost)
    if i >= M or j >= N or cost < 0
        return 0
    if i == M - 1 and j == N - 1
        if cost == mat[i][j] then return 1
        else return 0
    remaining = cost - mat[i][j]
    down = countPaths(mat, i + 1, j, remaining)
```

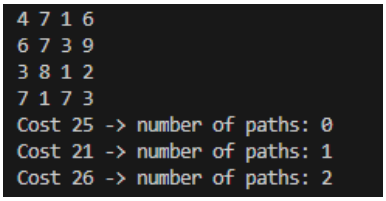
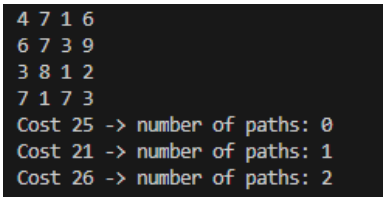
```
right = countPaths(mat, i, j + 1, remaining)
return down + right
```

Solving by hand

The smallest path is 4 + 7 + 1 + 3 + 1 + 2 + 3 = 21. Two paths have a cost of 26: 4 + 7 + 1 + 3 + 1 + 7 + 3 and 4 + 6 + 7 + 3 + 1 + 2 + 3.

All 20 path costs: 21, 26, 26, 27, 27, 29, 31, 31, 31, 32, 32, 32, 32, 32, 34, 35, 36, 36, 37, 37.

None of them is 25, so the answer by hand for cost 25 is 0 paths.

	Acala	**Esteta**
Screenshot		
Analysis	I used recursion to count the paths by subtracting the cost of each cell while moving right or down. For the matrix in the activity, none of the 20 paths has a cost of 25, so the answer is 0. My hand solution and the code gave the same result.	By using recursion to count the number of paths and summing up the cost value of the path it took. In the matrix there is no path that has an exact cost of 25 which matches the handle solution. Therefore my answer is 0
Conclusion	I learned the different algorithmic strategies: recursion, brute force, backtracking, greedy, divide and conquer, and dynamic programming. Greedy does not always give the best answer, like in n = 10 where greedy gives 4 steps but the best is 3. Dynamic programming saves the answers of the subproblems so they are not solved again.	I learned of the different algorithm strategies Recursion brute force back tracking greedy and divide and conquer. During the activity I realized that these algorithm strategies use algorithms that we have already used before. Where Greedy simply follows a local algorithm which means it does not always provide the optimal global solution.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

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I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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